Table A3. Detectable Exoplanet Surfaces and Considerations for Interpretations

Surface Type (1)	Table A2 Categories (2)	Potential Considerations for Interpretation if Detected (3)
Ultramafic Surface	Ferropericlase, Olivine, Pyroxene, Peridotite, Pyroxenite	According to Solar System theory, once a planet has formed its initial surface will be molten and, as it cools, the first material to condense out are dense ultramafic materials that sink forming the upper mantle. In the modern-day Solar System, ultramafic mantle material is partially melted to form mafic (basaltic) material on all rocky planets, though solid ultramafic rocks can be brought to the surface as large inclusions in other rocks (xenoliths) or via dislocated lithospheric slabs (see Peridotite in Table A2). Hu et al. (2012) argue that an ultramafic surface can be an indication of primary crust with mantle overturn, or a secondary crust formed from hot lavas. Mantle overturn is the process wherein largescale convection causes the mantle to essentially 'flip' so that dense material rises to top and light mantle material sinks, which could have occurred in the past on Mars, the Earth, the Moon (Elkins-Tanton et al. 2005; Bédard 2018; Liang et al. 2024). The hot lavas mentioned in Hu et al. (2012) are formed by some process like tidal heating, that completely melts ultramafic material (as opposed to partially melting it) in the subsurface and transports the melt to the surface to cool), which is analogous to the ultramafic lavas of Io (Breuer et al. 2022). An ultramafic surface could also be the remnant of a planet's initial formation chemical composition (very high Mg/Si ratio), the result of stripping away of the crust (and revealing the mantle), or dayside reprocessing from intense incident irradiation.
Mafic Surface	**Basalt**, Trachybasalt, Basaltic Andesite, Tephrite, Phonolite, Gabbro, Anorthosite, Andesite (Intermediate), Trachyte (Intermediate), Basalts in Table A5	From exoplanet mantle modeling and Solar System observations, basalts (and therefore, mafic rocks) are generally thought to be the most common surface component on rocky planets. This is because the secondary crusts of most planets will be basaltic as a natural consequence of what crystallizes (controlled by phase equilibria) from partial melts of ultramafic mantle material (which, from the modeling of rocky exoplanet compositions derived from host star compositions (e.g., Putirka & Rarick 2019), is expected to be the dominate mode of exoplanet materials). Additionally, most Solar System resurfacing processes (see 'Resurfacing' in Table A1) and models for tidally-locked exoplanets (e.g., terminator recycling in Meier et al. 2026) result in mafic surfaces. However, if a planet orbits a Mg-poor star (and therefore can have a more exotic mantle composition, as probed in the Putirka & Xu (2021) white-dwarf pollution study), a mantle could be more mafic (instead of ultramafic), and a mafic surface could indicate processes similar to the ones listed for 'Ultramafic Surfaces'. In Hu et al. (2012), it is argued that a feldspathic surface (i.e., anorthosite) is indicative of a primary crust without mantle overturn (where as the magma ocean cools rocks, like anorthosite, float to the top) while a basaltic surface is indicative of the formation of a a secondary crust. If a mafic rock-type (like basalt) is detected alongside a more felsic rock-type (such as andesite or granite), it can indicate that there are processes such as the remelting of basalts (which occurs due to plate tectonics in subduction zones on Earth) or extensive fractional crystallization (a process which forms felsic material from a mafic parental melt on Earth, Mars, and the Moon) (Udry et al. 2018). Modelers can test for specific Earth-like geologic processes by comparing their data to the basalt library of First et al. (2025).

Table A3 *continued*

Table A3 *(continued)*

Surface Type (1)	Table A2 Categories (2)	Potential Considerations for Interpretation if Detected (3)
Felsic Surface	Silica, Feldspar, Andesite (Intermediate), Trachyte (Intermediate), **Granite**, Rhyolite	In the Solar System, felsic rocks only compose a large surface component percentage of the Earth, making up the core of continental crust. The buildup of felsic continental crust (over geologic timescales) on Earth has been attributed to the partial melting of mafic, oceanic crust at subduction zones that generate silica-rich magmas (e.g., Gazel et al. 2015). Hu et al. (2012) argue that granitic surfaces are indicators of a tertiary crust when assuming an Earth-like mantle. However, felsic surfaces are not always an indicator of past or present plate tectonics since exoplanets can form initially silicte-rich, non Solar System-like mantles (Putirka & Xu 2021). If a planet's mantle is mafic or felsic, partial melting of the mantle will produce silica-rich magma without plate tectonics. Additionally, metamorphic processes (i.e., rocks subject to high temperatures and high pressures) can form rocks with granite-like compositions (i.e. gneiss) that then can be uncovered on the surface via erosion or active bombardment.
Oxidized Surface	Hematite	If a planet has an oxidizing atmosphere, any iron on the surface will be oxidized. Hematite is just one common example of oxidized iron. In Hu et al. (2012), hematite is indicative of oxidative weathering. Curiously, hematite was recently discovered on the Moon, despite its reducing and thin atmosphere (Li et al. 2020), though this has been attributed to mass transport of oxygen ions from the Earth to the Moon (Zeng et al. 2025). An oxidized surface might also be related to the interior oxygen fugacity of the object (see Oxygen Fugacity in Table A1).
Reduced Surface	Pyrite	In the solar-system, pyrite is not common as a bulk surface component since it rapidly decomposes when exposed to an oxidizing atmosphere. Therefore, if was detected on an exoplanet it would potentially indicate a largely reducing environment that keeps pyrite stable. In Hu et al. (2012), it is argued that pyrite ('metal-rich') is indicative of a primary crust with its mantle ripped off, meaning the planet has lost a substantial amount of its crustal and mantle material either due to large impacts (which are predicted to be the cause of the high density population of super-Earths (Cambioni et al. 2025)), extreme crustal evaporation via incident irradiation (see example, K2-22 b (Tusay et al. 2025), or remnant planetary cores around white dwarfs (Veras & Wolszczan 2019)). A reducing surface might also be related to the interior oxygen fugacity of the object (see Oxygen Fugacity in Table A1).
Magma Surfaces	Glasses in Table A6	Some tidally locked, rocky exoplanets (like 55 Cancri e) are expected to be hot enough that their daysides host, either full or partial, magma oceans. The composition of these magma oceans can be determined spectrally from their instrinic emission (see Fortin et al. (2022, 2024)), thin vaporized rock atmospheres (Schaefer & Fegley 2009; Miguel et al. 2011; Ito et al. 2015; Kite et al. 2016; Teske et al. 2025), or thick outgassed atmospheres (Hu et al. 2024) (see Oxygen Fugacity in Table A1). Drawing geologic inferences from the top-level composition of a magma ocean require a framework that includes planetary properties (like gravity), temporal properties (age of planet), atmospheric properties, and water content (see useful review, Chao et al. 2021). Whether or not a magma ocean does exist on the dayside of a rocky planet, and the magmas ability to dissolve or outgas volatiles, is strong function of incident irradiation recieved, the crustal composition (as different rocks have different melting temperatures and volatile storage properties), and oxygen fugacity (Brugman et al. 2021; Gaillard et al. 2021; Lin et al. 2021; Meier et al. 2026). If solid crustal material exists surrounding a partial magma ocean, it might be subject to contact metamorphism from neighboring magma, or low-pressure high-temperature from incident irradiation (see Metamorphic Rocks in Table A1).

Table A3 *continued*

Table A3 *(continued)*

Surface Type (1)	Table A2 Categories (2)	Potential Considerations for Interpretation if Detected (3)
Aqueously Altered Surface	Clays, Salts, some Basalts in Table A5	While many known rocky exoplanets are expected to be too hot on their tidally-locked, observable daysides to harbour surface liquid water, water can alter the surface and leave remnants indicating its past existence. Clays only form on Earth from the interaction of phyllosilicates (like feldspars) and liquid water. Salts are often an indicator of past or present water in the Solar System (i.e., on Earth, Mars, Europa, etc), where they are left as a residue on the surface from evaporating water. Salts are not expected to be stable at high (>1700K) temperatures that can exist on the daysides of tidally locked rocky planets, though salt glaciers have been proposed to exist on shadowed regions of Mercury (Rodriguez et al. 2023). Specific minerals, like amphibole and serpentine, can be present in rocks and can indicate that the rock either formed from magma that had large amounts of dissolved water, or the rock interacted with external water, sometime in its past (First et al. 2025).

Note—Here, we further contextualize the geological categories in Table A2 into broad, detectable surface types for rocky exoplanets in the era of JWST, and provide potential pathways to interpret each one if detected. Similar to Tables A1 and A2, this table is not extensive and was generated to help guide future considerations for exogeology studies and foster collaboration between geologists and exoplanet scientists. Basalt and Granite in boldface due to their extensive use in this paper. (1) Surface type category, (2) which categories from Table A2 fall under each surface type, and (3) potential considerations for how to geologically interpret a surface if detected.

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